Ningzhe Shi

Ph.D. Candidate in Computer Science and Technology

shiningzhe21b@ict.ac.cn Beijing, China TrainByte

Research Profile

Ph.D. candidate researching 6G intelligent networks and AI service orchestration, with a focus on communication, sensing, computing, and knowledge resource optimization through reinforcement learning and optimization methods.

Education

Ph.D. Candidate **Institute of Computing Technology, Chinese Academy of Sciences; University of Chinese Academy of Sciences**, Computer Science and Technology — Beijing, China; Sept 2021 – present
- Ph.D. advisor: Prof. Yiqing Zhou
- Discipline rated A+ in the national subject assessment

B.Eng. **Xidian University**, Communication Engineering — Xi'an, China; Sept 2017 – July 2021
- Undergraduate mentor: Prof. Junyu Liu
- Comprehensive recommendation ranking: 1st in class
- Discipline rated A+ in the national subject assessment

Internship

China Mobile, Digital Intelligence Department, AI Agent Algorithm Intern — China; Jan 2026 – Mar 2026

Conducted research on long-horizon memory mechanisms for AI agents.
- Studied retrieval-augmented memory, hierarchical storage, memory compression, forgetting, and dynamic updating.
- Reproduced and analyzed the Mem-alpha architecture and modeled memory maintenance as a policy optimization problem.
- Investigated memory-aware task scheduling in cloud-edge collaborative systems.

Research Experience

Institute of Computing Technology, Chinese Academy of Sciences, Lead Researcher, Service Satisfaction-Aware MEC Resource Allocation — Dec 2022 – Dec 2024
- Jointly optimized user selection, subchannel scheduling, power allocation, and edge-computing resources in NOMA-based multi-cell MEC networks.
- Achieved a gap of less than 1% from exhaustive-search performance and improved average user service satisfaction by more than 50%.

Institute of Computing Technology, Chinese Academy of Sciences, Lead Researcher, ISAC-Driven AIGC Resource Allocation — Jan 2024 – June 2025
- Built a content-accuracy-and-quality-aware model for sensing, computing, and communication resource allocation in ISAC-driven AIGC networks.
- Designed an LP-guided deep reinforcement learning framework.
- Improved the content-accuracy-and-quality-aware metric by more than 10% over learning baselines and by more than 50% over generation-quality-only schemes.

Institute of Computing Technology, Chinese Academy of Sciences, Lead Researcher, Long-Term Fairness-Aware Wireless Scheduling — May 2022 – May 2024

- Modeled imperfect channel-state information and long-term alpha-fair utility for wireless communication systems.
- Designed a reward-adaptive multi-agent reinforcement-learning framework.
- Improved long-term fairness by more than 13% while increasing the overall service rate by more than 10%.

Institute of Computing Technology, Chinese Academy of Sciences, Task Leader and Technical Lead, UAV-Assisted Secure Communications — Sept 2021 – June 2023

- Jointly optimized UAV positioning, mmWave MIMO beamforming, deceptive signal design, and power allocation.
- Achieved a performance gap of no more than 3% from the global optimum, improved secure transmission rate by more than 9% over direct-link transmission, and by more than 100% over the no-deception scheme.

Institute of Computing Technology, Chinese Academy of Sciences, Technical Lead, Full-Duplex Ad Hoc Networks — Oct 2022 – June 2023

- Designed multi-channel interference coordination and MAC-layer access mechanisms for in-band full-duplex ad hoc networks.
- Built a system-level simulation platform and produced an authorized invention patent and a registered software copyright.

Publications

Service Satisfaction Based User Selection and Resource Allocation for NOMA-Based Multi-Cell MEC Networks — 2026
Ningzhe Shi, Yiqing Zhou, Ling Liu, Yihao Wu, Hanxiao Yu, Jinglin Shi
10.1109/TMC.2026.3673357 (IEEE Transactions on Mobile Computing)

Content Accuracy and Quality Aware Resource Allocation Based on LP-Guided DRL for ISAC-Driven AIGC Networks — 2026
Ningzhe Shi, Yiqing Zhou, Ling Liu, Jinglin Shi, Yihao Wu, Haiwei Shi, Hanxiao Yu
10.1109/TMC.2026.3652144 (IEEE Transactions on Mobile Computing)

Long-Term Fairness From Real-Time Decisions: Reward Adaptive Multi-Agent DRL for Wireless Communication With Imperfect CSI — 2026
Ningzhe Shi, Ling Liu, Yiqing Zhou, Hanxiao Yu, Yihao Wu, Jingya Yang, Jinglin Shi
10.1109/TCCN.2026.3692166 (IEEE Transactions on Cognitive Communications and Networking)

Deception-Based Secure Communication for UAV-Assisted mmWave MIMO Systems — 2026
Ningzhe Shi, Yiqing Zhou, Yu Zhang, Zhijun Han, Ling Liu, Jinglin Shi
10.23919/JCC.fa.2025-0337.202604 (China Communications)

Intellectual Property

抗跟踪式干扰的信号发送方法、信号接收方法及系统 — Sept 2024
Granted invention patent: CN202211134854.X

一种基于中继位置选择和功率分配的抗干扰方法及系统 — May 2025
Granted invention patent: CN202310078808.0

无线网络资源分配系统的构建方法和资源管理方法 — Apr 2026
Granted invention patent: CN202310354794.0

无线网络资源分配系统的构建方法和资源管理方法 Apr 2023
Patent application: PCT/CN2023/090283

一种全双工自组织网络实时仿真系统及方法 Dec 2023
Patent application: CN202311719290.0

一种多用户 MIMO 信号检测模型 Jan 2024
Patent application: CN202410018551.4

一种面向自组织网络的信道接入方法和通信系统 Mar 2024
Patent application: CN202410348337.5

一种移动自组网数据路由方法、工业自组网系统 June 2025
Patent application: CN202510873622.3

一种环境和资源状态感知的资源分配方法和系统 July 2025
Patent application: CN202510955830.8

基于集成感知与通信的 AI 生成内容服务的资源分配方法 July 2025
Patent application: CN202511053332.0

联合信道估计与信号检测的方法、信号接收机及存储介质 Oct 2025
Patent application: CN202511536493.5

一种面向高速移动通信的发射装置及语义传输方法 Mar 2026
Patent application: CN202610406457.5

大规模全双工网络仿真系统 Jan 2024
Registered software copyright: 2024SR0082117

Honors and Awards

Outstanding Graduate Student, Xinghang Program June 2026
Journal of Electronics & Information Technology; national selection of 22 graduate students

Outstanding Poster Award June 2026
Frontiers in Electronics and Information Conference

UCAS Outstanding Student Model June 2026
University-wide top 1%

ICT Outstanding Student Jan 2026
Institute of Computing Technology / State Key Laboratory of Processors

E Fund Financial Technology Doctoral Scholarship Jan 2025
12 recipients institute-wide

UCAS Outstanding Student Cadre May 2025

UCAS Outstanding Student June 2023

UCAS First-Class Scholarship 2021–2025
Awarded for five consecutive years

Outstanding Graduate, Xidian University June 2021

First-Class Graduation Scholarship, Xidian University June 2021

Honorable Mention, Mathematical Contest in Modeling 2020

National Scholarship 2019

National Second Prize, China Undergraduate Mathematical Contest in Modeling 2019

Third Prize, CETC-10 Communication Cup 2019

Skills

Modeling and Optimization
Wireless and computing resource scheduling; ISAC; AI-service orchestration; convex optimization; linear programming; heuristic optimization

Artificial Intelligence
Deep reinforcement learning; multi-agent reinforcement learning; generative models; diffusion models; quantum reinforcement learning

Programming and Simulation
Python; C++; MATLAB; Shell; PyTorch; custom reinforcement-learning environments; system-level network simulation

Academic Writing
IEEE journal writing; technical reports; literature review; simulation-result analysis

Academic Service

Program Committee

- Technical Program Committee Member, IEEE International Conference on Communications (IEEE ICC) 2026

Academic Correspondence

- Academic Correspondent, Digital Communications and Networks

Peer Review

- IEEE Transactions on Mobile Computing
- IEEE Transactions on Communications
- IEEE Internet of Things Journal
- IEEE Transactions on Vehicular Technology
- Science China Information Sciences
- China Communications
- IEEE International Conference on Communications
- IEEE Vehicular Technology Conference
- 通信学报
- 电子与信息学报

Outreach and Public Engagement

Science outreach volunteer, Chinese Academy of Sciences Public Science Day

Participated in public science communication and research outreach activities

Committee Member, UCAS Science Popularization Association

Leadership

Vice President, Haidian Volunteer Federation Branch

Youth League Branch Secretary and Volunteer Team Leader, Xidian University School of Telecommunications Engineering